



Pharmacological evidence that Ca^{2+} channels and, to a lesser extent, K^{+} channels mediate the relaxation of testosterone in the canine basilar artery

Martha B. Ramírez-Rosas^a, Luis E. Cobos-Puc^a, Enriqueta Muñoz-Islas^{a,1},
Abimael González-Hernández^a, Araceli Sánchez-López^a, Carlos M. Villalón^a,
Antoinette MaassenVanDenBrink^b, David Centurión^{a,*}

^a Departamento de Farmacobiología, Cinvestav-Coapa, Czda. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, C.P. 14330, México D.F., Mexico

^b Division of Vascular Medicine and Pharmacology, Department of Internal Medicine, Erasmus MC, Rotterdam, P.O. Box 2040, 3000 CA Rotterdam, The Netherlands

ARTICLE INFO

Article history:

Received 16 October 2010

Received in revised form 1 December 2010

Accepted 18 December 2010

Available online 28 December 2010

Keywords:

Ca^{2+} channels

Canine basilar artery

K^{+} channels

Non-genomic mechanisms

Testosterone

Vasorelaxation

ABSTRACT

Testosterone induces vasorelaxation through non-genomic mechanisms in several isolated blood vessels, but no study has reported its effects on the canine basilar artery, an important artery implicated in cerebral vasospasm. Hence, this study has investigated the mechanisms involved in testosterone-induced relaxation of the canine basilar artery. For this purpose, the vasorelaxant effects of testosterone were evaluated in KCl- and/or $\text{PGF}_{2\alpha}$ -precontracted arterial rings *in vitro* in the absence or presence of several antagonists/inhibitors/blockers; the effect of testosterone on the contractile responses to CaCl_2 was also determined. Testosterone (10–180 μM) produced concentration-dependent relaxations of KCl- or $\text{PGF}_{2\alpha}$ -precontracted arterial rings which were: (i) unaffected by flutamide (10 μM), DL-aminoglutethimide (10 μM), actinomycin D (10 μM), cycloheximide (10 μM), SQ 22,536 (100 μM) or ODQ (30 μM); and (ii) significantly attenuated by the blockers 4-aminopyridine (K_V ; 1 mM), BaCl_2 (K_{IR} ; 30 μM), iberiotoxin ($\text{BK}_{\text{Ca}^{2+}}$; 20 nM), but not by glybenclamide (K_{ATP} ; 10 μM). In addition, testosterone (31, 56 and 180 μM) and nifedipine (0.01–1 μM) produced a concentration-dependent blockade of the contraction to CaCl_2 (10 μM to 10 mM) in arterial rings depolarized by 60 mM KCl. These results, taken together, show that testosterone relaxes the canine basilar artery mainly by blockade of voltage-dependent Ca^{2+} channels and, to a lesser extent, by activation of K^{+} channels (K_{IR} , K_V and $\text{BK}_{\text{Ca}^{2+}}$). This effect does not involve genomic mechanisms, production of cAMP/cGMP or the conversion of testosterone to 17 β -estradiol.

© 2010 Elsevier Inc. All rights reserved.

1. Introduction

Androgens are known to affect the vascular tone, but their effects are diverse and depend on several factors including the concentration used, species, blood vessel under study and the experimental conditions. Some studies have reported that low concentrations of testosterone: (i) quickly increased coronary resistance and blocked adenosine-mediated vasodilatation in the perfused rat heart [1]; (ii) acutely inhibited bradykinin-induced relaxation in porcine coronary artery rings [2]; (iii) enhanced the

contractile responses to different agonists in the porcine coronary artery [3]; and (iv) induced endothelium-dependent relaxations in the rat mesenteric bed [4]. Furthermore, high concentrations of testosterone can produce direct (endothelium-independent) vasodilatation in a variety of vascular beds, an action mediated by a non-genomic pathway [5]. For instance, testosterone can relax directly the rat thoracic aorta [6,7], human radial [8] and umbilical [9,10] arteries, as well as rabbit [11] and pig [12,13] coronary arteries. Moreover, in men with coronary artery disease, acute i.v. administration of testosterone at supra-physiological concentrations (2.3 mg during 10 min) enhanced endothelium-dependent flow-mediated brachial artery reactivity which, in turn, produced vasodilatation [14]. These data suggest that testosterone may be beneficial to the cardiovascular system of men with coronary heart disease [15].

The mechanisms involved in the direct vasorelaxant effects of testosterone are still under debate and appear to be mediated via activation of K^{+} channels and/or blockade of Ca^{2+} channels, although steroid membrane receptors and cellular signalling pathways could also be involved [16]. Indeed, sex steroids can activate cyclic nucleotide signalling pathways [12,17–19].

Abbreviations: AC, adenylyl cyclase; cAMP, 3',5'-cyclic adenosine monophosphate; cGMP, 3',5'-cyclic guanosine monophosphate; DMSO, dimethyl sulfoxide; GC, guanylyl cyclase; ODQ, 1H-[1,2,4]oxadiazolo[4,3-a]quinoxalin-1-one; SNP, sodium nitroprusside; SQ 22,536, [9-(tetrahydro-2-furanyl)-9H-purine-6-amine].

* Corresponding author. Tel.: +52 55 5483 2866; fax: +52 55 5483 2863.

E-mail address: dcenturi@cinvestav.mx (D. Centurión).

URL: <http://www.cinvestav.mx/farmacobiologia/PersonalAcademico/cpd.html> (D. Centurión).

¹ Current address: Instituto Nacional de Perinatología, Montes Urales 800, Col. Lomas de Virreyes, Deleg. Miguel Hidalgo, C.P. 11000, México D.F., Mexico.

Even though the vasculature in general and cerebral blood vessels in particular are targets for sex steroids [20], to the best of our knowledge the effects of testosterone on the canine basilar artery and the mechanisms involved have not yet been reported. The basilar artery is important in the pathophysiology of cerebral vasospasm and it is one of the best models of experimental subarachnoid haemorrhage [21]. Hence, this study set out to investigate the effects of testosterone on the canine basilar artery and the mechanisms involved in these effects. Our results indicate that testosterone induces basilar relaxation through non-genomic pathways that involve: (i) blockade of Ca^{2+} influx through inhibition of voltage-dependent Ca^{2+} channels; and (ii) to a lesser extent, activation of inward rectifier K^+ channels (K_{IR}), voltage-sensitive K^+ channels (K_{V}) and large-conductance Ca^{2+} -activated K^+ channels ($\text{BK}_{\text{Ca}^{2+}}$), which are present in arterial smooth muscle [22].

2. Materials and methods

2.1. Tissue preparation

Eighteen male mongrel dogs were anaesthetized with sodium pentobarbitone (30 mg/kg, i.v.) and sacrificed by ex-sanguination from the common carotid artery. Then, the basilar artery was removed and rapidly placed in Krebs–Ringer solution of the following composition (mM): NaHCO_3 (24.9), NaCl (119.5), KCl (4.74), KH_2PO_4 (1.18), MgSO_4 (1.18), CaCl_2 (2.5) and glucose (12.0). The artery was cleaned of fat, blood and connective tissue and cut into 9 rings of 3 mm length. The endothelial layer was systematically removed by gentle rubbing with a stainless steel hook and the rings were horizontally mounted in organ bath chambers containing 10 ml Krebs solution at 37°C . The solution was continuously bubbled with 95% O_2 –5% CO_2 , resulting in a pH of 7.4. Changes in arterial tension were recorded isometrically by a force–displacement transducer (FT03, Grass Instrument Co., Quincy, MA, U.S.A.) connected to a data acquisition unit (MP 100; Biopac Systems Inc., Goleta, CA, U.S.A.). The rings were equilibrated for 1 h under a resting tension of 60 mN (6.0 g) (obtained from previous studies), before starting the experiments. It must be emphasised that the 9 basilar artery rings obtained from each animal received different treatments.

2.2. Relaxant effects of testosterone on the precontracted canine basilar artery

After an equilibration period of 60 min, the contractile responses to 60 mM KCl were elicited three times every 20 min. Then, the basilar rings were precontracted with 5-hydroxytryptamine (5-HT, 1 μM) and the effect of acetylcholine (ACh, 1 μM) was determined. Those rings in which ACh produced relaxation were excluded from the protocol. Subsequently, the rings were equilibrated for 40 min (with washing periods of 10 min each) and precontracted with 60 mM KCl (same procedure for the protocols mentioned below). When the contractile response had reached a plateau, a cumulative concentration–response curve to testosterone was established from 10 μM to 180 μM in 0.25 log increments (10 min each) in the absence or presence of vehicle or the antagonists/inhibitors/blockers. In vehicle-control experiments, ethanol alone was added cumulatively following the same scheme as that of testosterone administration and the maximum concentration reached by ethanol in the organ bath was 0.6% v/v.

2.3. Effect of antagonists/inhibitors/blockers on the relaxant response to testosterone

2.3.1. Aromatase and genomic pathways

To examine the involvement of testosterone receptors, aromatase (which converts testosterone to 17β -estradiol), tran-

scription and protein synthesis, in another set of experiments the contractile response to KCl was determined; then, the rings were incubated for 30 min with vehicle, flutamide (10 μM), aminoglutethimide (10 μM), actinomycin D (10 μM) or cycloheximide (10 μM). Then, a concentration–response curve to testosterone (10–180 μM) was obtained. The concentrations of androgen receptor antagonist [23] and transcription inhibitor [24] were high enough to completely block their corresponding mechanisms. The inhibitors of aromatase and protein synthesis were added at concentrations previously reported in similar *in vitro* studies [2,9,25].

2.3.2. cAMP and cGMP production pathways

Arterial rings precontracted with 3 μM $\text{PGF}_{2\alpha}$ were prepared to determine: (i) the relaxant effect to sodium nitroprusside (SNP; 1 nM to 10 μM) or testosterone (10–180 μM) in the presence of the guanylyl cyclase inhibitor ODQ (30 μM) or its vehicle (dimethyl sulfoxide; DMSO 0.1% v/v); and (ii) the relaxant effect to testosterone (10–180 μM) in the presence of vehicle (DMSO 0.1% v/v) or the adenylyl cyclase inhibitor, SQ 22,536 (100 μM).

2.3.3. K^+ channels

To analyze the possible involvement of K^+ channels in the vasorelaxant response to testosterone, the arterial rings precontracted by KCl were incubated independently for 30 min with glybenclamide (10 μM ; a selective inhibitor of K_{ATP} channels); 4-aminopyridine (1 mM; an inhibitor of K_{V} channels); BaCl_2 (30 μM ; an inhibitor of K_{IR} channel); and iberiotoxin (20 nM; a highly selective inhibitor of $\text{BK}_{\text{Ca}^{2+}}$ channels). After incubation, a cumulative concentration–response curve to testosterone (10–180 μM) was established. The concentration used of each inhibitor of K^+ channels was high enough to selectively block those channels in arterial smooth muscle according to Nelson and Quayle [22].

2.4. Effect of testosterone or nifedipine on the contractile responses to CaCl_2

To evaluate the capability of testosterone to block Ca^{2+} entry, concentration–response curves to CaCl_2 (10 μM to 10 mM) were obtained in the absence and in the presence of different concentrations of testosterone (18, 56 and 180 μM) and of the voltage-dependent Ca^{2+} channel blocker nifedipine (0.01–1 μM). For this purpose, arterial basilar rings without endothelium were incubated and adapted in Ca^{2+} -free Krebs solution containing 2 mM ethylenediaminetetraacetic acid (EDTA disodium salt) for 40 min; the rings were washed three times (10 min each) and subsequently incubated in Ca^{2+} -free high- K^+ (60 mM) depolarizing solution. Then, cumulative concentration–response curves to CaCl_2 were obtained. Later on, the arterial rings were washed with Ca^{2+} -free Krebs solution; when vascular tone reached baseline values, the rings were allowed to equilibrate for 40 min. During this period, the rings were washed three times (10 min each) and subsequently depolarized with Ca^{2+} -free high- K^+ (60 mM); 10 min later, each preparation was incubated for 30 min individually with vehicle (ethanol), testosterone or nifedipine. Finally, a cumulative concentration–response curve to CaCl_2 was constructed.

All animal procedures and the protocols of the present investigation were approved by our Institutional Ethics Committee (Cicual-Cinvestav) and followed the regulations established by the Mexican Official Norm for the Use and Welfare of Laboratory Animals (NOM-062-ZOO-1999) in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals in U.S.A.

2.5. Drugs

Apart from the anaesthetic (sodium pentobarbitone), the compounds used in this study were: 17 β -hydroxyandrost-4-en-3-one (testosterone), flutamide, actinomycin D, cycloheximide, tetraethylammonium chloride, glybenclamide and nifedipine (dissolved in absolute ethanol); DL-aminoglutethimide (dissolved in methanol); BaCl₂, iberiotoxin, CaCl₂, 4-aminopyridine, sodium nitroprusside and PGF_{2 α} (dissolved in bidistilled water). Finally [9-(tetrahydro-2furanyl)-9H-purine-6-amine] (SQ 22,536) and 1H-[1,2,4]oxadiazolo[4,3-a]quinoxalin-1-one (ODQ) were dissolved in DMSO. All of the above compounds were purchased from Sigma Chemical Co. (St. Louis, MO, U.S.A.). Each compound was added in aliquots of 10 μ l in the bath chambers of 10 ml.

2.6. Statistical analysis

The relaxant responses elicited by testosterone were expressed as percentage of the maximum contraction induced by either 60 mM KCl or 3 μ M PGF_{2 α} (100%). The contractile responses to CaCl₂ were expressed as percentage of the previous response to 60 mM KCl. All responses were expressed as mean $\pm$ standard error of the mean (s.e.m.) and the number of animals is represented by *n*. A one-way analysis of variance was used to compare the relaxant effects of different concentrations of testosterone, and a two-way analysis of variance was performed to compare the effects of testosterone against the antagonists/inhibitors/blockers, followed by the Student–Newman–Keuls' test. Statistical significance was accepted at *P* < 0.05.

3. Results

3.1. Relaxant responses to testosterone on the precontracted canine basilar artery

The basilar precontraction by 60 mM KCl resulted in a mean tension of 49.7 ± 0.8 mN (*n* = 18 dogs). Testosterone (10–180 μ M), but not vehicle (ethanol) (*n* = 6 rings obtained from 6 different basilar arteries each[†]), induced a concentration-dependent relaxation (*P* < 0.05) (Fig. 1). This relaxant response started a few seconds after each concentration of testosterone was added (not shown). The maximum response reached by 180 μ M testosterone was $112 \pm 2\%$,

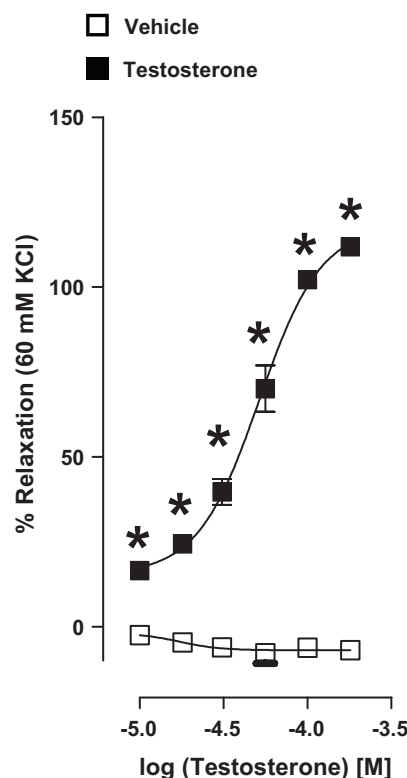


Fig. 1. Relaxant response to testosterone (10–180 μ M) in basilar rings without endothelium precontracted with 60 mM KCl. Cumulative concentration–response curves of testosterone (■) or vehicle (ethanol) (□) were constructed. Each point represents the mean $\pm$ s.e.m. (*n* = 6[†]). **P* < 0.05 vs vehicle.

and the effective concentration fifty (EC₅₀) of testosterone was 52 ± 1 μ M (*n* = 6, as previously described[†]). In contrast, cumulative additions of vehicle failed to significantly modify the basal contraction to KCl.

3.2. Effect of flutamide, aminoglutethimide, actinomycin D or cycloheximide on the relaxant response to testosterone

As shown in Fig. 2, the vasorelaxant response to testosterone was not significantly inhibited by flutamide (10 μ M,

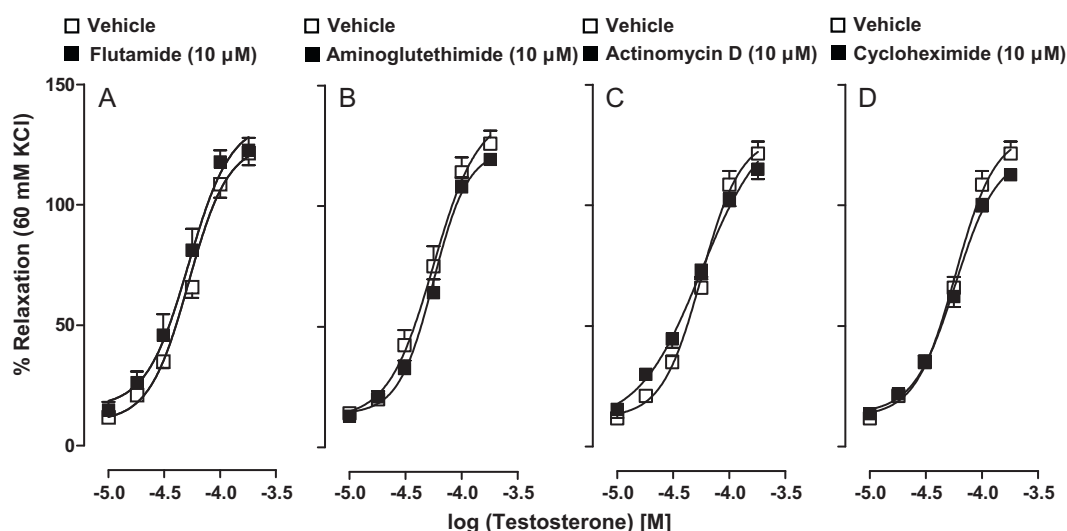


Fig. 2. Effects of pre-incubation of the canine basilar artery with: (A) the selective androgen receptor antagonist flutamide (10 μ M); (B) the aromatase inhibitor aminoglutethimide (10 μ M); (C) the transcription inhibitor actinomycin D (10 μ M); and (D) the protein synthesis inhibitor cycloheximide (10 μ M) on the relaxant response to testosterone. Each point represents the mean $\pm$ s.e.m. (*n* = 6[†]).

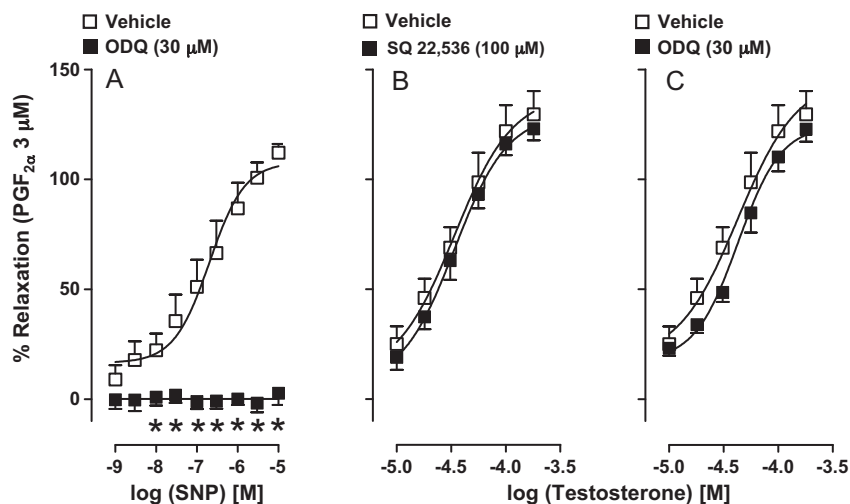


Fig. 3. Analysis of the effects of several compounds or their corresponding vehicle (in all cases DMSO 0.1% v/v □) on canine basilar artery rings precontracted with PGF_{2α} (3 μM) and relaxed by: (A) sodium nitroprusside (SNP) in the presence of ODQ (■); or testosterone in the presence of either (B) SQ 22,536 (■) or (C) ODQ (■). Each point represents the mean $\pm$ s.e.m. ($n = 6$). * $P < 0.05$ vs vehicle.

an androgen receptor antagonist), aminoglutethimide (10 μM; an aromatase inhibitor), actinomycin D (10 μM; a transcription inhibitor) or cycloheximide (10 μM; a protein synthesis inhibitor).

3.3. Effect of SQ 22,536 or ODQ on the relaxant response to testosterone

The relaxant response to testosterone (10–180 μM) was not significantly inhibited by 100 μM SQ 22,536 or 30 μM ODQ (see Fig. 3B and C, respectively). It is noteworthy that 30 μM ODQ was high enough to abolish the relaxation to SNP (1 nM to 10 μM) in the canine basilar artery (see Fig. 3A).

3.4. Effect of testosterone on K⁺ channels

To establish the role of K⁺ channels in the relaxant response to testosterone, several specific inhibitors were used. The cumulative concentration–response curve to testosterone was significantly

blocked by either 4-aminopyridine (1 mM; K_V inhibitor) or BaCl₂ (30 μM; K_{IR} inhibitor) (particularly at 56 and 100 μM testosterone). Indeed, the relaxation to 56 and 100 μM testosterone in the presence of vehicle ($86 \pm 7\%$ and $118 \pm 4\%$) was significantly reduced by 1 mM 4-aminopyridine ($^{a}70 \pm 6\%$ and $^{a}108 \pm 2\%$, respectively) or 30 μM BaCl₂ ($^{a}61 \pm 6\%$ and $^{a}102 \pm 4\%$, respectively) ($^{a}P < 0.05$ vs vehicle). In addition, the relaxation to 31, 56 and 100 μM testosterone in the presence of vehicle ($46 \pm 4\%$, $86 \pm 7\%$ and $118 \pm 4\%$, respectively) was significantly inhibited by 20 nM iberiotoxin (a BK_{Ca2+} inhibitor; $^{a}33 \pm 3\%$, $^{a}60 \pm 6\%$ and $^{a}103 \pm 6\%$) ($^{a}P < 0.05$ vs vehicle). In contrast, the relaxation to testosterone (10–180 μM) was not significantly modified by 10 μM glybenclamide (a K_{ATP} inhibitor) (see Fig. 4).

3.5. Effect of testosterone on the CaCl₂ concentration–response curve

CaCl₂ (10 μM to 10 mM) induced concentration-dependent contractions on basilar artery rings depolarized by KCl 60 mM in

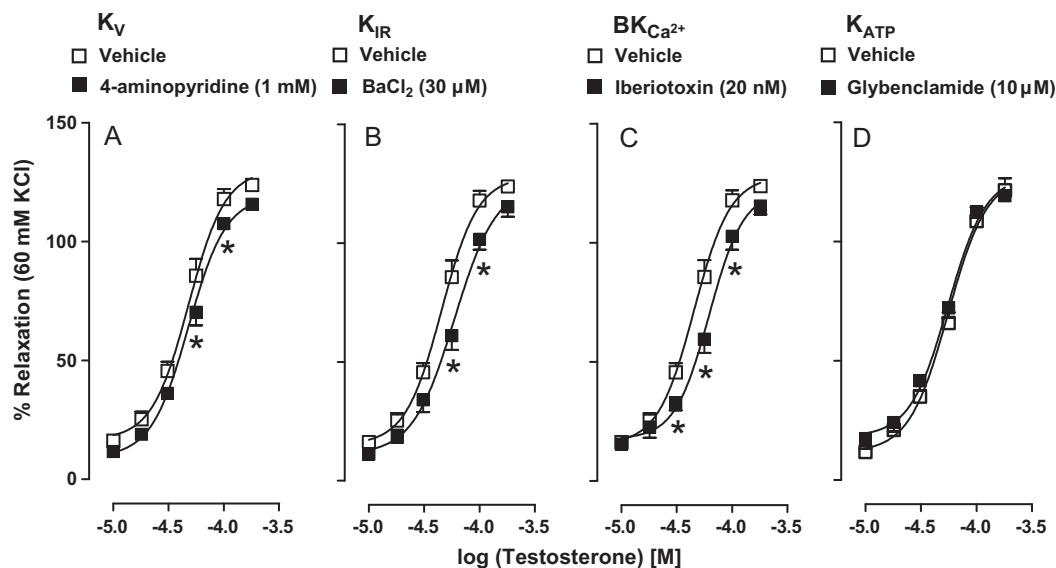


Fig. 4. Cumulative concentration–response curves to testosterone (10–180 μM) obtained in canine basilar artery rings precontracted with 60 mM KCl in the presence of: (A) 4-aminopyridine; (B) BaCl₂; (C) iberiotoxin; or (D) glybenclamide. Each point represents the mean $\pm$ s.e.m. ($n = 6$). * $P < 0.05$ vs vehicle.

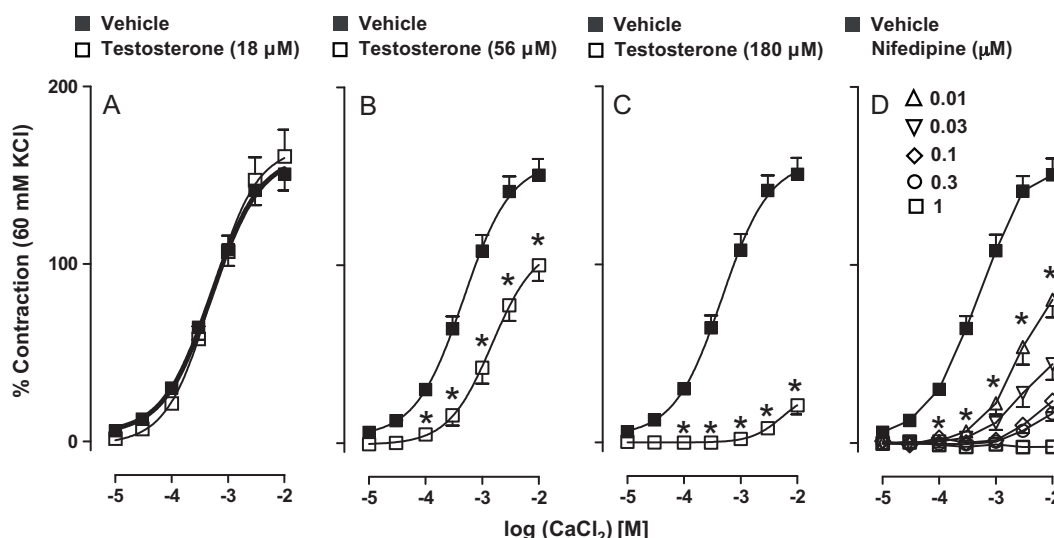


Fig. 5. Effects of testosterone (18, 56 and 180 μM) or nifedipine (0.01–1 μM) on the cumulative concentration–response curves to CaCl_2 in Ca^{2+} -free depolarizing solution (60 mM KCl) on canine basilar artery rings. Results are expressed as mean $\pm$ s.e.m. ($n = 6$). * $P < 0.05$ vs vehicle.

Ca^{2+} -free solution. Pre-incubation with testosterone (18, 56 and 180 μM) or nifedipine (0.01–1 μM) significantly attenuated the contraction to CaCl_2 in a concentration-dependent manner (see Fig. 5).

4. Discussion

4.1. No evidence for the involvement of genomic mechanisms, conversion to 17 β -estradiol or cAMP/cGMP production in the relaxant response to testosterone

Since the relaxant response to testosterone was elicited within a few seconds of its application, a rapid non-genomic mechanism may be involved, as reported in porcine coronary [2] and prostatic small [25] arteries. This is strengthened by the failure of actinomycin D, cycloheximide or flutamide to modify the response to testosterone.

In addition, the role of aromatase (which is expressed in vascular smooth muscle [26]) can be excluded as 10 μM aminoglutethimide failed to inhibit the response to testosterone. Thus, a direct vasorelaxant response is implied, as previously shown in rabbit coronary arteries [27], rat mesenteric bed [4] and pig prostatic small arteries [25].

Alternatively, the non-genomic effect of testosterone could involve: (i) an increase in cAMP as reported in denuded rat aortic rings [28]; and (ii) the sex hormone binding globulin receptor (SHBG-R) which, being coupled to Gs-protein [29,30], may increase the production of cAMP and activate protein kinase A [31,32]. Therefore, the possible role of cAMP/cGMP pathways was evaluated by using $\text{PGF}_{2\alpha}$ as a contractile agent since its mechanism of action includes activation of G protein-coupled receptors. The failure of the enzyme inhibitors SQ 22,536 (AC; [33]) and ODQ (GC; [34]) to inhibit the response to testosterone excludes the role of cAMP/cGMP pathways. Indeed, 30 μM ODQ was enough to abolish the relaxation to SNP (Fig. 3A) and 100 μM SQ 22,536 abolished the rise in cAMP induced by iloprost in the isolated guinea-pig aorta [35].

4.2. Possible involvement of K^+ channels

Arterial smooth muscle contains four distinct types of K^+ channels, namely: $\text{BK}_{\text{Ca}^{2+}}$, K_V , K_{ATP} and K_{IR} [22]. To evaluate the possible

involvement of these channels, specific blockers were used. The fact that the relaxation to testosterone was significantly blocked by 4-aminopyridine (K_V), BaCl_2 (K_{IR}) or iberiotoxin ($\text{BK}_{\text{Ca}^{2+}}$), but not by glibenclamide (K_{ATP}), suggests that K_V , K_{IR} , and $\text{BK}_{\text{Ca}^{2+}}$, but not K_{ATP} , channels are involved. Admittedly, the slight blockade produced by these compounds implies that the above channels are partly involved, although K^+ channels do not participate in the relaxant response to testosterone in other blood vessels [25,36]. In agreement with our results, the relaxation to testosterone is mediated by: (i) K_V channels in aortic rings from spontaneously hypertensive [37] and Sprague–Dawley [7] rats as well as in rabbit coronary artery strips [11]; and (ii) $\text{BK}_{\text{Ca}^{2+}}$ channels in human umbilical artery [10] and rat mesenteric arterial bed and aorta [4,38].

4.3. Role of Ca^{2+} channels

Vascular smooth muscle can express low and high voltage-activated Ca^{2+} channels; however, the most abundant high voltage-activated Ca^{2+} channel of vascular smooth muscle is the L-type $\alpha_{1\text{C}}$ Ca^{2+} channel, which provides an influx of Ca^{2+} for vasoconstriction during membrane depolarization [39]. With this in mind, our results show that testosterone and nifedipine (L-type voltage dependent Ca^{2+} channel blocker) blocked the contraction to CaCl_2 in Ca^{2+} -free and depolarized medium. Under these conditions, the contraction to CaCl_2 is due to an increase in $[\text{Ca}^{2+}]_i$ through influx of Ca^{2+} by voltage-dependent calcium channels. Therefore, we could suggest that both testosterone and nifedipine can block L-type voltage-dependent calcium channels sensitive to dihydropyridine. Similar findings have been reported in porcine prostatic [25] and coronary [13,40] arteries as well as in rat aorta [38] and coronary [41] arteries. Furthermore, Nikitina et al. [42] showed that the canine basilar artery smooth muscle expresses functional L- and T-type voltage-dependent Ca^{2+} channels. Accordingly, lower concentrations of testosterone block the major α_1 subunit of the L-type Ca^{2+} channel whereas higher concentrations inhibit T-type Ca^{2+} channels [43,44]. These findings suggest that the basilar relaxation to testosterone is mainly mediated by a blockade of Ca^{2+} influx through the inhibition of voltage-dependent Ca^{2+} channels. Besides, the highest concentration of testosterone induced a vasorelaxant effect greater than 100% KCl (Fig. 1), implying that it is affecting the myogenic tone, as previously reported

[38]; thus voltage-dependent Ca^{2+} channels are partly involved [45].

4.4. Further validation of the concentrations of inhibitors/antagonist/blockers used

It is noteworthy that $0.1 \mu\text{M}$ flutamide exhibited potent antian-drogenic activity [46], whereas $10 \mu\text{M}$ flutamide was enough to completely block androgen receptors [23] and to abolish the transcriptional effect regulated by androgen receptors in human aortic smooth muscle cells [47]. Moreover, actinomycin D (0.005 – $0.08 \mu\text{g/ml}$ equivalent to 0.003 – $0.63 \mu\text{M}$) inhibited RNA synthesis [24] and at $5 \mu\text{g/ml}$ ($3.98 \mu\text{M}$) blocked the relaxation to 5α -dihydrotestosterone in rat uterus [48]. Interestingly, $1 \mu\text{g/ml}$ ($3.5 \mu\text{M}$) cycloheximide abolished testosterone-induced inhibition on testosterone biosynthesis in Leydig cells, an event linked to protein synthesis [49]. Hence, the failure of $10 \mu\text{M}$ cycloheximide to affect testosterone-induced vasodilatation suggests that protein synthesis is not involved, although we do not have a positive control for this concentration; unfortunately, higher concentrations of cycloheximide significantly increased vascular tone (data not shown). Thus, we cannot categorically exclude the possible role of protein synthesis in testosterone-induced vasodilatation in our study. In addition, the concentrations used of K^+ channel blockers are enough to block their respective channels [22]. Although it is tempting to suggest that higher concentrations of K^+ channel blockers could produce a higher blockade of the relaxation to testosterone, the use of higher concentrations of these blockers significantly modified the baseline contractile responses to KCl (data not shown).

Admittedly, the concentrations of testosterone used in our study are in the pharmacological (μM) rather than in the physiological (nM) range, as shown in previous *in vitro* reports [6,8,41]. Although these concentrations are non-physiological, treatment with supra-physiological i.v. doses of testosterone in men with coronary artery disease induced a marked brachial vasodilatation [14]. Moreover, testosterone therapy in men with angina [15] and heart failure [50,51] have shown positive results. Notwithstanding, the concentrations used *in vitro* do not necessarily correlate with those used *in vivo*. Hence, the concentrations in organ baths are higher because testosterone (a non-polar molecule) has to cross different tissue layers for finally interacting with vascular smooth muscle into the aqueous environment; in contrast, under *in vivo* conditions, the target tissue is reached directly through the blood circulation.

In conclusion, our results in the canine basilar artery suggest that testosterone-induced vasorelaxation is: (i) unrelated to genomic or cAMP/cGMP pathways, its conversion to 17β -estradiol, or activation of androgen receptors; and (ii) mainly mediated by blockade of voltage-dependent Ca^{2+} channels and, to a lesser extent, by activation of K_V , K_IR , and $\text{BK}_{\text{Ca}^{2+}}$ channels.

Acknowledgement

The authors would like to express their gratitude to CONACyT (México) for their financial support.

References

- [1] Figueroa-Valverde L, Luna H, Castillo-Henkel C, Munoz-Garcia O, Morato-Cartagena T, Ceballos-Reyes G. Synthesis and evaluation of the cardiovascular effects of two, membrane impermeant, macromolecular complexes of dextran–testosterone. *Steroids* 2002;67:611–9.
- [2] Teoh H, Quan A, Man R. Acute impairment of relaxation by low levels of testosterone in porcine coronary arteries. *Cardiovasc Res* 2000;45:1010–8.
- [3] Teoh H, Quan A, Leung S, Man R. Differential effects of 17β -estradiol and testosterone on the contractile responses of porcine coronary arteries. *Br J Pharmacol* 2000;129:1301–8.
- [4] Tep-areenan P, Kendall DA, Randall MD. Testosterone-induced vasorelaxation in the rat mesenteric arterial bed is mediated predominantly via potassium channels. *Br J Pharmacol* 2002;135:735–40.
- [5] Jones RD, Pugh PJ, Jones TH, Channer KS. The vasodilatory action of testosterone: a potassium-channel opening or a calcium antagonistic action? *Br J Pharmacol* 2003;138:733–44.
- [6] Costarella CE, Stallone JN, Rutecki GW, Whittier FC. Testosterone causes direct relaxation of rat thoracic aorta. *J Pharmacol Exp Ther* 1996;277:34–9.
- [7] Ding AQ, Stallone JN. Testosterone-induced relaxation of rat aorta is androgen structure specific and involves K^+ channel activation. *J Appl Physiol* 2001;91:2742–50.
- [8] Seyrek M, Yildiz O, Ulusoy HB, Yildirim V. Testosterone relaxes isolated human radial artery by potassium channel opening action. *J Pharmacol Sci* 2007;103:309–16.
- [9] Perusquía M, Navarrete E, González L, Villalón CM. The modulatory role of androgens and progestins in the induction of vasorelaxation in human umbilical artery. *Life Sci* 2007;81:993–1002.
- [10] Cairrão E, Álvarez E, Santos-Silva AJ, Verde I. Potassium channels are involved in testosterone-induced vasorelaxation of human umbilical artery. *Naunyn Schmiedeberg's Arch Pharmacol* 2008;376:375–83.
- [11] Won E, Won J, Kwon S, Lee Y, Nam T, Ahn D. Testosterone causes simultaneous decrease of $[\text{Ca}^{2+}]_i$ and tension in rabbit coronary arteries: by opening voltage dependent potassium channels. *Yonsei Med J* 2003;44:1027–33.
- [12] Deenadayalu VP, White RE, Stallones JN, Gao X, García AJ. Testosterone relaxes coronary arteries by opening the large-conductance, calcium-activated potassium channel. *Am J Physiol Heart Circ Physiol* 2001;281:H1720–7.
- [13] Crews JK, Khalil RA. Antagonistic effects of 17β -estradiol, progesterone, and testosterone on Ca^{2+} entry mechanisms of coronary vasoconstriction. *Arterioscler Thromb Vasc Biol* 1999;19:1034–40.
- [14] Ong PJ, Patrizi G, Chong WC, Webb CM, Hayward CS, Collins P. Testosterone enhances flow-mediated brachial artery reactivity in men with coronary artery disease. *Am J Cardiol* 2000;85:269–72.
- [15] Webb CM, McNeill JG, Hayward CS, de Zeigler D, Collins P. Effects of testosterone on coronary vasomotor regulation in men with coronary heart disease. *Circulation* 1999;100:1690–6.
- [16] Michels G, Hoppe UC. Rapid actions of androgens. *Front Neuroendocrinol* 2008;29:182–98.
- [17] Mügge A, Riedel M, Barton M, Kuhn M, Lichtlen PR. Endothelium independent relaxation of human coronary arteries by 17β -estradiol *in vitro*. *Cardiovasc Res* 1993;27:1939–42.
- [18] White RE, Darkow DJ, Lang JLF. Estrogen relaxes coronary arteries by opening BK_{Ca} channels through a cGMP-dependent mechanism. *Circ Res* 1995;77:937–42.
- [19] Rodriguez J, Garcia de Boto MJ, Hidalgo A. Mechanism involved in the relaxant effect of estrogens on rat aorta strips. *Life Sci* 1996;58:607–15.
- [20] Orshal JM, Khalil RA. Gender, sex hormones, and vascular tone. *Am J Physiol Regul Integr Comp Physiol* 2004;286:R233–49.
- [21] Megyesi JF, Vollrath B, Cook DA, Findlay JM. *In vivo* animal models of cerebral vasospasm: a review. *Neurosurgery* 2000;46:448–90.
- [22] Nelson MT, Quayle JM. Physiological roles and properties of potassium channels in arterial smooth muscle. *Am J Physiol* 1995;268:C799–822.
- [23] Nazareth LV, Weigel NL. Activation of the human androgen receptor through a protein kinase A signaling pathway. *J Biol Chem* 1996;271:19900–7.
- [24] Perry RP, Kelley DE. Inhibition of RNA synthesis by actinomycin D: characteristic dose–response of different RNA species. *J Cell Physiol* 1970;76:127–39.
- [25] Navarro-Dorado J, Orensanz LM, Recio P, Bustamante S, Benedito S, Martínez AC, et al. Mechanisms involved in testosterone-induced vasodilation in pig prostatic small arteries. *Life Sci* 2008;83:569–73.
- [26] Mendelsohn ME, Rosano GMC. Hormonal regulation of normal vascular tone in males. *Circ Res* 2003;93:1142–5.
- [27] Yue P, Chatterjee K, Beale C, Poole-Wilson PA, Collins P. Testosterone relaxes rabbit coronary arteries and aorta. *Circulation* 1995;91:1154–60.
- [28] Montañón LM, Calixto E, Figueroa A, Flores-Soto E, Carbajal V, Perusquía M. Relaxation of androgens on rat thoracic aorta: testosterone concentration dependent agonist/antagonist L-type Ca^{2+} channel activity, and 5β -dihydrotestosterone restricted to L-type Ca^{2+} channel blockade. *Endocrinology* 2008;149:2517–26.
- [29] Nakhla AM, Leonard J, Hryb DJ, Rosner W. Sex hormone-binding globulin receptor signal transduction proceeds via a G protein. *Steroids* 1999;64:213–6.
- [30] Rosner W, Hryb DJ, Khan MS, Nakhla AM, Romas NA. Androgen and estrogen signaling at the cell membrane via G-proteins and cyclic adenosine monophosphate. *Steroids* 1999;64:100–6.
- [31] Heinlein CA, Chang C. The roles of androgen receptors and androgen-binding proteins in nongenomic androgen actions. *Mol Endocrinol* 2002;16:2181–7.
- [32] Waldkirch E, Ückert S, Schultheiss D, Geismar U, Bruns C, Scheller F, et al. Non-genomic effects of androgens on isolated human vascular and nonvascular penile erectile tissue. *BJU Int* 2007;101:71–5.
- [33] Lippe GT, Ardizzone C. Actions of vasopressin and isoprenaline on the ionic transport across the isolated frog skin in the presence and the absence of adenylyl cyclase inhibitors MDL 12330A and SQ22536. *Comp Biochem Physiol C* 1991;99:209–11.
- [34] Garthwaite J, Southam E, Boulton CL, Nielsen EB, Schmidt K, Mayer B. Potent and selective inhibition of nitric oxide-sensitive guanylyl cyclase by 1H-[1,2,4]oxadiazolo[4,3-a]quinoxalin-1-one. *Mol Pharmacol* 1995;48:184–8.

- [35] Turcato S, Clapp LH. Effects of the adenylyl cyclase inhibitor SQ 22,536 on iloprost-induced vasorelaxation and cyclic AMP elevation in isolated guinea-pig aorta. *Br J Pharmacol* 1999;126:845–7.
- [36] Kim JK, Han WH, Lee MY, Myung SC, Kim SC, Kim MK. Testosterone relaxes rabbit seminal vesicle by calcium channel inhibition. *Korean J Physiol Pharmacol* 2008;12:73–7.
- [37] Honda H, Unemoto T, Kogo H. Different mechanisms for testosterone-induced relaxation of aorta between normotensive and spontaneously hypertensive rats. *Hypertension* 1999;34:1232–6.
- [38] Tep-areenan P, Kendall DA, Randall MD. Mechanisms of vasorelaxation to testosterone in the rat aorta. *Eur J Pharmacol* 2003;465:125–32.
- [39] Lacinová L, Hofmann F. Voltage-dependent calcium channels. In: Sperelakis N, Kurachi Y, Terzic A, Cohen MV, editors. *Heart physiology and pathophysiology*. San Diego, CA: Academic Press; 2001. p. 247–57.
- [40] Murphy JG, Khalil RA. Decreased $[Ca^{2+}]_i$ during inhibition of coronary smooth muscle contraction by 17 β -estradiol, progesterone, and testosterone. *J Pharmacol Exp Ther* 1999;291:44–52.
- [41] Jones RD, English KM, Jones TH, Channer KS. Testosterone-induced coronary vasodilatation occurs via a non-genomic mechanism: evidence of a direct calcium antagonism action. *Clin Sci* 2004;107:149–58.
- [42] Nikitina E, Zhang ZD, Kawashima A, Jahromi BS, Bouryi VA, Takahashi M, et al. Voltage-dependent calcium channels of dog basilar artery. *J Physiol* 2007;580:523–41.
- [43] Scragg JL, Jones RD, Channer KS, Jones TH, Peers C. Testosterone is a potent inhibitor of L-type Ca^{2+} channels. *Biochem Biophys Res Commun* 2004;318:503–6.
- [44] Hall J, Jones RD, Jones TH, Channer KS, Peers C. Selective inhibition of L-type Ca^{2+} channels in A7r5 cells by physiological levels of testosterone. *Endocrinology* 2006;147:2675–80.
- [45] Davis MJ, Hill MA. Signaling mechanisms underlying the vascular myogenic response. *Physiol Rev* 1999;79:387–423.
- [46] Xu L, Sun H, Chen J, Bian Q, Quian J, Song L, et al. Evaluation of androgen receptor transcriptional activities of bisphenol A, octylphenol and nonylphenol in vitro. *Toxicology* 2005;216:197–203.
- [47] Son B, Akishita M, Lijima K, Ogawa S, Maemura K, Yu J, et al. Androgen receptor-dependent transactivation of growth arrest-specific gene 6 mediates inhibitory effects of testosterone on vascular calcification. *J Biol Chem* 2010;285:7537–44.
- [48] Aparicio Sánchez JA, Gutierrez M, Hidalgo A, Cantabrana B. Effects of androgens on isolated rat uterus. *Life Sci* 1993;53:269–74.
- [49] Fanjul LF, González J, Quintana J, Santana P, Hernández I, Cabrera J, et al. Inhibition of steroidogenesis in neonatal Leydig cells by unknown factor(s) present in spent media of androgen-treated cultured testicular cells from adult rats. *J Androl* 1993;14:419–27.
- [50] Pugh PJ, Jones RD, West JN, Jones TH, Channer KS. Testosterone treatment for men with chronic heart failure. *Heart* 2004;90:446–7.
- [51] Malkin CJ, Jones TH, Channer KS. Testosterone in chronic heart failure. *Front Horm Res* 2009;37:183–96.